

BDS-SMC2 UAV Rescue System

Publication Plan — Two Papers (Rev. 3)

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Updated 14 June 2026 — figures finalised, Q1 readiness, Paper 2 publishability + experiment plan

1. Publication Strategy

Revision 2 consolidates the encoding results (Gaps 1 + 6) into Paper 1: the 112-bit rescue payload is now the carrying contribution and belongs with the reliability and latency evidence that proves it deliverable. Paper 2 becomes the end-to-end system paper (RTK injection, GCS decode, ROS 2, UAV mission pipeline), matching the group integration now in place.

■ **Positioning note: the BDS-3 GSMC system paper (Li et al., *Advances in Space Research* 67(5):1701-1708, 2021; 2149 TX, 97.72%) already field-measured the link. Our claim is therefore the **NARROWED** one below — never claim 'first field measurement of BDS-3 SMC'.**

Paper	Scope	Target Venue	Core Claim
Paper 1	Gaps 1, 2, 3, 6 + 112-bit rescue payload	MDPI Drones (primary) / Sensors / IEEE Access	First environment-stratified characterisation of BDS-3 SMC carrying a complete 6-field rescue payload: 112 bits, delivery reliability $\geq 93.7\%$ (Wilson 95% LB), 2.57 s mean latency, integrated into a UAV mission pipeline
Paper 2	End-to-end system: RTK injection, portal GCS, ROS 2, MAVLink	IEEE Access / MDPI Drones	Full survivor-to-UAV pipeline: RTK coordinate $\rightarrow$ 112-bit BDS-SMC $\rightarrow$ portal $\rightarrow$ decode $\rightarrow$ ROS 2 mission trigger, with stage-by-stage latency budget

2. Paper 1 — Transmission Layer (Gaps 1, 2, 3, 6 + 112-bit payload)

Field	Detail
Working title	A 112-bit Binary Rescue Payload over BeiDou-3 Short Message Communication: Environment-Stratified Reliability and Latency for UAV Search-and-Rescue
Venue	MDPI Drones (primary — publishes this genre, fast review); fallbacks: MDPI Sensors, IEEE Access
Submission	After 1 remaining field day (Gap 2 midday + evening, 112-bit verification TX) — target August 2026
Status	Empirical work ~75% complete. Defence sections drafted (BDS_SMC2_Paper1_Sections.md): comparison table, emulated-RTK framing, Wilson claim, exclusion appendix, repeat-TX policy.

2.1 Data Status (audited 2026-06-12)

Dataset	Requirement	Collected	Status
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Gap 3 reliability	4 environments x 3 locations x ~20 TX	232 valid TX (61/57/57/57)	COMPLETE
Gap 2 latency	3 time-of-day sessions x 30 TX, ALL on the 112-bit payload (Option B)	0 of 90 (30-TX ASCII baseline archived as secondary comparison)	NEED morning + midday + evening
Gap 1 / Gap 6 encoding	Bit counts + round-trip verification	Software-verified incl. 112-bit on lab T001-T006	COMPLETE
112-bit hardware proof	1 TX accepted by BDS module + portal receipt	0	NEED 1 test TX

2.2 Key Results to Report (verified against raw data, 2026-06-12)

<i>Metric</i>	<i>Value</i>	<i>Source</i>
Delivery success (valid TX)	232/232; report as Wilson 95% LB ≥93.7% per env, ≥98.4% pooled	gap3_field_test.csv
Environment effect	None detectable: chi-sq=0.000, df=3, p=1.000; Fisher pairwise all 1.0	gap3_analysis.py
Mean TX latency (T1→T3)	ASCII baseline: 2574.5 ms (n=30, sd 1094.7) — archived; operational-payload sessions TBD	gap2_latency_ascii_baseline.csv
ANOVA time-of-day	TBD after 3 sessions on 112-bit payload (Option B); ASCII-vs-binary secondary comparison via --ascii-baseline	gap2_analysis.py
Gap 1 encoding	ASCII 264 bits vs 112-bit rescue payload (-57.6%)	decode_binary.py round-trip
Gap 6 telemetry	ASCII 368 / Huffman 184 / Binary 112 bits (-69.6% vs ASCII; 39% under Huffman with MORE fields)	gap6_telemetry.csv
Payload completeness	All 6 lab rescue-report fields (T001-T006) round-trip bit-perfect at 7dp	decode_binary.py test
BDS-3 headroom	98 bits under the 210-bit regional limit	design
Excluded records	33 rows, single cause (T3 instrumentation bug, OS-1, re-collected) — full appendix drafted	Paper1_Sections C.2

2.3 Remaining Task List — Paper 1

#	<i>Task</i>	<i>When</i>	<i>Tool / Ref</i>
1	Flash MODE 1 firmware once; Gap 2 morning session (30 TX) — TX #1 doubles as the 112-bit acceptance check + on-air bit-accounting check on the portal	Field day, 08:00-10:00	run_gap2_morning.bat
2	Gap 2 midday session (30 TX) + optional power measurement	Field day, 12:00-14:00	run_gap2_midday.bat
3	Gap 2 evening session (30 TX)	Field day, after 18:00	run_gap2_evening.bat

4	Activate portal reader (browser tokens) + run dashboard for portal-corroborated delivery evidence	Field day	portal_reader.py / tx_dashboard.py
5	Optional: power draw per TX (USB meter)	Field day	J/message metric
6	Run ANOVA across 3 sessions; regenerate figures	After field day	gap2_analysis.py / generate_figures.py
7	Fill [CITE] placeholders in drafted sections (LoRaWAN, COSPAS-SARSAT, Iridium, BDS-3 GSMC paper)	This week	BDS_SMC2_Paper1_Sections.md
8	Assemble full draft: drafted sections + results + figures	After analysis	venue template
9	Supervisor review of submission-ready draft (not an outline)	Before submission	email
10	Submit	Target Aug 2026	venue portal

2.5 Figures (6 — the canonical set)

Fig	File	Section	Shows
1	payload_route_illustration.png	III	End-to-end signal path + instrumentation
2	gap1_encoding_comparison.png	V	ASCII 264 vs binary 112 bits (-57.6%)
3	gap6_telemetry_comparison.png	V	368 / 184 / 112 bits + 210-bit limit
4	gap3_success_rate.png	VI	Delivery by environment + Wilson CIs
5	gap3_location_breakdown.png	VI	12 locations, 232 TX
6	fig_gap2_cdf.png	V	Latency CDF (regenerate after field day)

Ignore old/duplicate figures (fig2_*, fig3_*, _pres*, fig_*summary, fig_research_overview) — not the Paper 1 set.

2.6 Q1 Readiness

Criterion	Standing	Gap to close
Novelty	STRONG — first environment-stratified field measurement; binary beats Huffman	Position vs Li et al. [R1]
Rigor	STRONG — Wilson CIs, ANOVA, chi-sq, published raw data + exclusion log	Finish Gap 2 ANOVA (field day)
Significance	GOOD — infrastructure-free, triage-complete rescue signalling	Comparison table sells it
Reproducibility	EXCELLENT — open stdlib toolchain, raw CSVs published	Add data-availability statement
Positioning	NEEDS WORK	Related-work + cite R1-R6 (writing, not experiment)
Presentation	IN PROGRESS	Clean figures (done) + tight writing

Realistic target: MDPI Drones (Q1/Q2, publishes this genre). IEEE TAES is true Q1 but the single-unit + emulated-RTK gaps would trigger major revisions. Blockers to Q1: (1) do the field day for real latency; (2) single hardware unit — disclose; (3) emulated RTK — defensible via Objective 5, closed in Paper 2.

3. Paper 2 — End-to-End Rescue Pipeline (system paper)

Field	Detail
Working title	End-to-End BDS-3 Short Message Rescue System: RTK Coordinate Injection, GCS Decoding, and UAV Mission Integration via ROS 2
Venue	IEEE Access or MDPI Drones
Prerequisite	Paper 1 submitted. Group integration validated (ROS 2 node + verify_integration.sh already in the team repo).
What it proves	Paper 1 proves the transmission layer; Paper 2 proves the full pipeline: RTK fix → encode → satellite → portal → decode → ROS 2 topic → UAV mission, with a latency budget per stage.
Note	Group stack is ROS 2 (not ROS 1 as in Rev. 1 of this plan) and uXRCE-DDS/px4_msgs (not MAVROS). EmergencyCoordinate.msg extension (alt / uncertainty / priority / survivor_id) pending group approval.

3.1 Experiments To Collect (after Paper 1)

Experiment	Min	Proves	Who/where
RTK→BDS accuracy chain	30/set	Chain preserves RTK precision (~1cm = quantization)	YOU — node+GCS
GCS decode latency	30+	Portal→decode stage timing	YOU — node+GCS
End-to-end latency budget *	30+ runs	Per-stage RTK→trigger timing — THE novel core	GROUP stack
Multi-survivor triage *	3x6 TX	Survivor-ID + priority → P0-first rescue	GROUP stack
ROS 2 publish rate	50+	Real-time capable	GROUP stack
Mission-trigger latency	20+	Topic publish → UAV reacts	GROUP stack
UAV positional-error model	derived	Drift = speed x latency (no new hardware)	YOU — analysis

* = must-have (the paper's core). All GROUP-stack rows are gated on ONE integrated session (Gazebo + ROS + your node running together). YOU-rows can be pre-collected alone before that session.

3.2 Publishability — confronted honestly

Hard truth	Consequence / mitigation
It is a simulation / proof-of-concept paper	UAV = Gazebo, RTK = emulated; only the satellite link is real. Publishable in Drones/Sensors/IEEE Access as 'simulation-validated architecture' — NOT a top field-robotics venue. Frame honestly.
Authorship boundary	Objectives 1-4 are group mates' work. Either a multi-author group paper (cleanest), or YOUR paper claiming only BDS+GCS+integration with their modules cited. Decide BEFORE writing.
Must not read as 'Paper 1 + glue'	Novel core MUST be the end-to-end latency budget + multi-survivor triage + operations layer (retry/recycle/self-heal), NOT 'we connected modules'.

Upgrade: one real flight	A single real UAV flying to one real BDS-transmitted coordinate converts it from simulation to demonstrated system and unlocks better venues. Highest value-per-effort addition.
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Verdict: publishable as a simulation-validated end-to-end system paper (Q1/Q2 applied venue) IF the integration data is collected, authorship is settled, and novelty is framed as above. Sequence: Paper 1 submitted FIRST; do not start Paper 2 writing until then.

3.3 Five Levers to Lift Paper 2 to Q1 (from YOUR side, no group dependency)

Lever	Action	When
1. Reframe	Title/subject = YOUR GCS + operations layer, UAV is the consumer. 'A self-healing, delivery-assured ground-control and operations layer for satellite rescue messaging, integrated with an autonomous UAV pipeline.'	now (writing)
2. Real, not sim	State plainly: BDS TX + satellite + portal + GCS + operations layer are validated on REAL hardware over the live BDS-3 link; only the UAV response is simulated. Strongest underused argument.	now (framing)
3. Rigour	(a) end-to-end latency budget = per-stage distribution + CIs + bottleneck; (b) black-box satellite transit dt distribution ($n \geq 100$, unreported elsewhere); (c) self-healing layer validated vs $1-(1-p)^k$ delivery-assurance model.	integration session + analysis
4. Artifact	Package the open stdlib GCS+operations toolchain (portal reader, decoder, dashboard, auto-sender+retry) as a reusable release. Reproducibility = Q1 differentiator; most BDS-SMC papers ship nothing.	now (packaging)
5. One real flight	Champion ONE real drone flight to one real BDS-transmitted coordinate. Converts the paper from 'simulation-validated' to 'demonstrated system'. You provide the coordinate; make it the headline ask.	group integration

Ceiling: Levers 1-4 alone -> solid Q1/Q2 (Drones/Sensors/IEEE Access). Add Lever 5 (real flight) -> genuine Q1. Items 1, 4 are pure writing/packaging doable NOW before any integration session.

4. Combined Timeline (Rev. 3)

Period	Activity	Output
Now	Draft Related Work + positioning; verify R5/R6; open PR; .msg meeting	Paper 1 positioning done
Field day	Gap 2 morning+midday+evening (112-bit, Option B); 112-bit acceptance; dashboard evidence	Paper 1 empirical record COMPLETE
After field day	gap2_analysis.py -> fill slots; regenerate Fig 6; assemble full draft	Submission-ready draft to supervisor
Then	Revise per supervisor; Ubuntu integration session with group	Paper 1 final; Paper 2 integration data starts
Target Aug 2026	Submit Paper 1 (MDPI Drones)	Under review
Integration session	Run full Obj1-5 stack together; collect latency budget + multi-survivor	Paper 2 must-have data
Sep-Nov 2026	Settle authorship; write Paper 2; (stretch) one real flight	Paper 2 draft -> submit (IEEE Access / Drones)

5. Project-Wide Strategy — How Many Papers from Obj 1-5

Honest count: ~5-6 papers possible across the team, but most objectives are APPLICATIONS of established methods (RTKLIB, PX4/QGC, vision detection, RRT*). Q1 rejects 'we applied known method X' — the publishable angle is the disaster-rescue gap the standard method does NOT cover. Realistic Q1 candidates: Obj 5 (yours) and Obj 1; the rest are Q2/Q3 or contributors to a joint paper.

Obj	Publishable argument (NOT the base method)	Realistic
1 RTK	RTK reliability under compound-disaster GNSS degradation + viability-gating for safe landing (their Level-3 total-failure scenarios ARE the novelty)	Q1/Q2
2 QGC	Autonomous precision-landing decision framework gated on real-time RTK viability	Q2/Q3 or joint
3 Detect	Detection-to-geolocation pipeline, OR a disaster-survivor dataset, OR detection under rubble/occlusion	Q2/Q3
4 Path	Uncertainty-aware planning to a survivor REGION defined by the rescue search-radius R (uses Obj 5's R field)	Q2/Q3
5 BDS	Field-measured BDS-SMC rescue link + complete 112-bit payload (Paper 1); end-to-end GCS system (Paper 2)	Q1 / Q1-Q2

5.1 Cross-Objective Papers (often stronger than the parts)

Combination	Why it is novel
Obj1 RTK viability -> Obj2 landing	Position-confidence-gated autonomous landing (closed safety loop)
Obj5 uncertainty R -> Obj4 planning	Uncertainty-aware planning to a survivor region, not a point
Obj3 detect -> Obj5 coord -> Obj4 path	Detection-confirmed rescue retargeting

Smartest strategy: NOT 5 separate papers — rather 1-2 strong solo papers (yours, Obj 1) + ONE joint flagship system paper giving everyone authorship. Quality over a forced paper each (echoes the supervisor's '100 unpublishable papers = 0').